Geo-SimCLR: Self-supervised Contrastive Pre-training of Sentinel-2 Representations for U.S. Corn Belt Yield Prediction

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Abstract

Open-source Sentinel-2 imagery has made satellite-based agricultural monitoring widely accessible, but representation learning remains the bottleneck: annual USDA county-year corn-yield labels are scarce relative to the abundance of unlabelled imagery. We introduce **Geo-SimCLR**, a SimCLR variant tailored to satellite time series in three respects: (i) positive pairs are the same field at adjacent biweekly timestamps rather than two augmented copies of one image; (ii) anchor draws are balanced across USDA Crop Reporting Districts so within-batch negatives are agro-climatically diverse on average; (iii) augmentations are restricted to the safe, satellite-aware subset (no spectral band dropout, no RGB colour jitter, no mixup). Trained from scratch on ∼645 k anchor pairs with an 11.2 M-parameter ResNet-18 backbone, Geo-SimCLR is the strongest single encoder under *both* temporal and spatial generalisation splits (test $R^2 = 0.669$ on the time split, 0.660 on the county split), beating a frozen DINOv2-base, a frozen Prithvi-EO 2.0 (300 M params), a supervised end-to-end ConvLSTM, and the strongest hand-crafted NDVI baseline. We also diagnose some *partial dimensional collapse* of the learned representation and show, through M-PHATE, plain PCA, and GradCAM, that the encoder reliably captures geographic identity but only weakly encodes seasonal phenology. We discuss concrete fixes – harder positive pairs, decorrelation regularisers, 4-channel AG+NDVI input, and partial fine-tuning – as future work.

1 Introduction

Precise, timely, and spatially-resolved estimates of agricultural yield matter for agricultural planning, food-security assessment, government policy and spending, monetary-policy decisions whose inputs include food-price inflation, and – as a more recent development – financial-market participants who use satellite-derived crop-condition signals to anticipate USDA reports [13, 21, 8]. The pre-eminent input variable in this literature is the Normalised Difference Vegetation Index (NDVI), a hand-crafted ratio of red and near-infrared surface reflectance which correlates with canopy chlorophyll content and, by extension, with end-of-season yield [15, 17].

The bottleneck in this pipeline is not data acquisition. The European Space Agency's Sentinel-2 mission has made medium-resolution multispectral imagery of the entire planet freely available with a 5-day revisit cadence [5]; CropNet [17] packages a biweekly, county-aligned subset for 2,291 U.S. counties from 2017–2022. The bottleneck is *representation*: handcrafted spectral indices like NDVI discard texture, spatial heterogeneity and intra-tile context, while training deep imagery models end-to-end on the few thousand annual labels per state overfits dramatically. The same gap exists in many adjacent Earth-observation problems – climate change attribution, deforestation tracking, biodiversity and habitat monitoring, urban growth, water-resource and drought analysis, disaster

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response – where labels are scarce but unlabelled imagery is plentiful. A general-purpose recipe for learning rich, transferable representations of multispectral satellite tiles would therefore have value far beyond corn-yield prediction; we use county-year corn yield as a tractable, label-rich case study.

We propose **Geo-SimCLR**, a contrastive self-supervised pre-training recipe tailored to satellite time series. Building on SimCLR [3], our central design choice is to replace SimCLR's augmentation-pair positives with *temporal* positive pairs – the same physical field at adjacent biweekly timestamps – so that the representation is encouraged to be invariant to short-term weather and view nuisance while remaining sensitive to canopy state. We additionally diversify within-batch negatives by sampling anchors with weights inversely proportional to their Crop Reporting District (CRD) frequency, and we restrict augmentations to a safe satellite-aware subset (no spectral dropout, no RGB-flavoured colour jitter, no mixup; see Section 3).

Contributions.

1. We design and train Geo-SimCLR, an 11.2 M-parameter ResNet-18 encoder from scratch on $\sim$645 k temporal positive pairs drawn from seven-state Corn Belt Sentinel-2 imagery.
2. We benchmark Geo-SimCLR against a strong NDVI baseline, two foundation models (frozen DINOv2-base [22] and frozen Prithvi-EO 2.0 [11]), a supervised end-to-end ConvLSTM [25], and a random-init ResNet-18 sanity baseline, under both temporal (train 2017–2020 / test 2022) and spatial (random 20 % counties held out) generalisation splits with bootstrap 95 % confidence intervals.
3. We report good spacial representation but also a partial dimensional collapse of the learned representation and visualise these through M-PHATE, plain PCA, and GradCAM, showing that the encoder captures geographic identity very cleanly but encodes seasonal phenology only weakly.
4. We propose a prioritised list of fixes – harder positives, decorrelation regularisers, 4-channel AG+NDVI input, partial fine-tuning, and head-to-head benchmarking against temporal-positive SSL prior art – and discuss the broader applicability of the recipe.

Reproducibility. Code and notebooks are available at `https://github.com/petkovip/crop-yields--satellite-image-representation`; a short project presentation video is at `https://youtu.be/G777uqAYG7Q`.

2 Background and related work

Hand-crafted spectral indices. Decades of remote-sensing-based yield prediction rest on indices such as NDVI ($(\mathrm{NIR} - \mathrm{Red})/(\mathrm{NIR} + \mathrm{Red})$), the soil-adjusted SAVI, and the enhanced EVI, fed into linear or tree-based regressors with political-unit fixed effects [26, 15]. These features summarise canopy greenness compactly but discard the spatial texture and within-field heterogeneity that the underlying multispectral imagery actually carries.

Supervised CNN/RNN models. A second wave applied convolutional and recurrent supervised networks to imagery time series. ConvLSTM [25] interleaves convolution inside an LSTM cell to consume gridded spatial-temporal data; Khaki and Wang's CNN-RNN framework [15] couples a CNN over satellite scenes with an RNN over biweeks but in practice operates on tabular soil and management features rather than imagery; Fan et al.'s GNN-RNN [6] adds a graph over neighbouring counties; and Lin et al.'s multi-modal MMST-ViT [17], packaged with the CropNet dataset we use here, fuses Sentinel-2 imagery with WRF-HRRR weather and a USDA crop modality. The CropNet authors report that most of MMST-ViT's gain comes from the weather modality: dropping WRF-HRRR raises corn RMSE by 7.4 bu/ac, more than twice the gap between any two visual encoders (17, Table 5). Among the imagery-only supervised baselines they benchmark, ConvLSTM is therefore the only one directly comparable to a frozen-encoder + linear-probe recipe at fixed input modality, and we re-implement it as our supervised imagery comparator.

Self-supervised representation learning. Contrastive self-supervised learning [3, 9] learns representations by pulling together augmentation-pair embeddings and pushing others apart, achieving near-supervised linear-probe performance on ImageNet without labels. DINOv2 [22] extends this to large ViT-based foundation models trained on 142 M curated natural RGB images and demonstrates strong out-of-the-box transfer. Masked autoencoders [10] provide an alternative reconstruction-based objective; SatMAE [4] and Prithvi-EO 2.0 [11] adapt the MAE recipe to multispectral satellite tiles. Closer to our design, the SeCo [18], GASSL [1], and GeoSSL families use temporal positive pairs from co-located satellite scenes; we did not benchmark against these specific implementations here – a clear future-work item – but acknowledge them as the closest prior art.

Representation diagnostics. A growing body of work examines failure modes of contrastive SSL. Jing et al. [12] formalise *dimensional collapse* in which the representation concentrates on a low-rank subspace of the embedding; Zbontar et al. [27] (Barlow Twins) and Bardes et al. [2] (VICReg) add explicit decorrelation terms that prevent it. We borrow the SVD-based effective-rank diagnostic of Roy and Vetterli [23] and the multislice trajectory visualiser M-PHATE of Gigante et al. [7], complemented by PHATE [20] and UMAP [19].

Choice of comparators in this work. Within the imagery-only frozen-feature regime, we compare Geo-SimCLR to (i) hand-crafted NDVI baselines fitted with LightGBM [14]; (ii) two strong frozen foundation models – DINOv2-base and Prithvi-EO 2.0 – chosen as the natural "what does a strong general-purpose / EO-specialised SSL model give you out of the box?" baselines; (iii) a random-init ResNet-18 sanity baseline; and (iv) the supervised end-to-end ConvLSTM described above. We do not chase MMST-ViT's multi-modal supervised RMSE, both because it is multi-modal and because the CropNet ablation attributes most of its gain to weather rather than vision. Comparison against the temporal-positive-pair SSL lineage (SeCo, GASSL, GeoSSL, SatMAE) is left to future work.

3 Methods

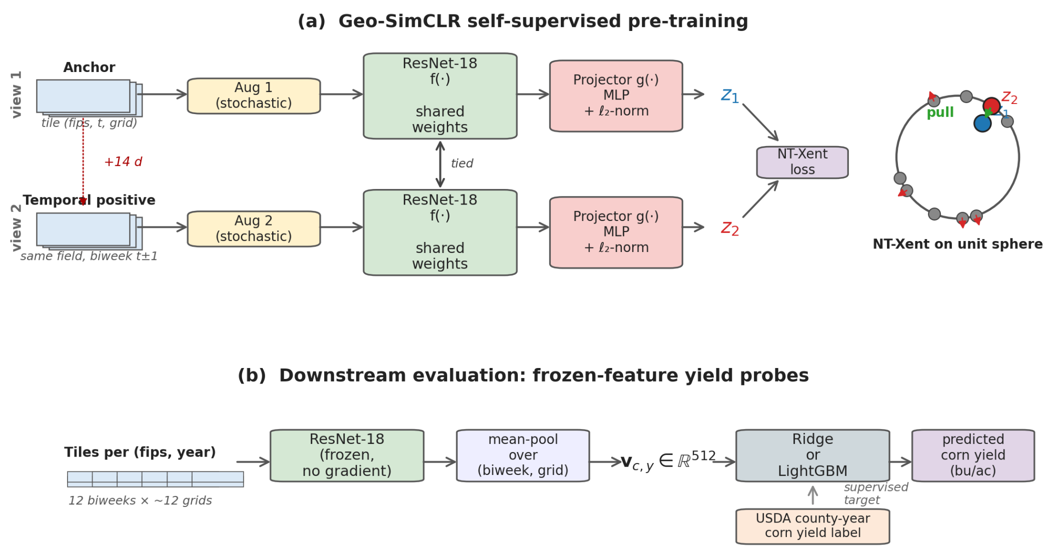

Figure 1: Geo-SimCLR pipeline. (a) Self-supervised pre-training: an anchor tile and its same-field temporal positive (biweek $t \pm 1$, $\sim$14 days apart) are independently augmented, encoded by a shared ResNet-18 with a tied projection head, and the two ℓ_2-normalised projections z_1, z_2 are pulled together while all other batch members act as negatives via NT-Xent. (b) Downstream evaluation: each county-year's tiles are passed through the frozen encoder, mean-pooled into a single 512-d county-year vector, and fed to a Ridge or LightGBM probe whose target is the USDA county-year corn yield.

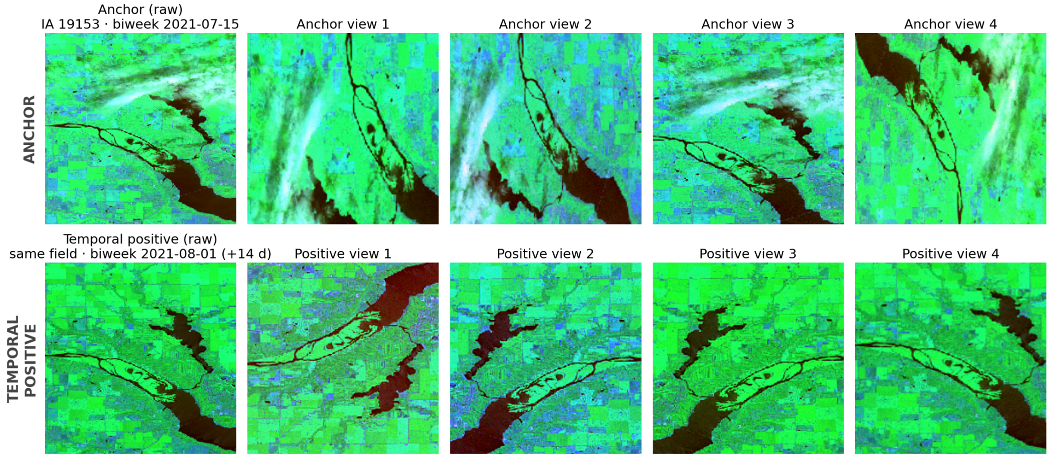

Figure 2: Geo-SimCLR pre-text task on real Sentinel-2 tiles, rendered in agriculture false-colour (SWIR1→R, NIR→G, Blue→B). Top row: an anchor tile (Polk County, IA, 2021-07-15) and four independent stochastic draws of the augmentation pipeline. Bottom row: the temporal positive (same county and grid index, 2021-08-01, $+14$ d) and four independent draws. The contrastive loss asks the encoder to make all eight augmented views similar to each other and dissimilar to other batch members. Augmentations are: random h/v flip, random rot90, random crop 160–208 px → resize 224, sharpness $\times 2.0$ with $p = 0.5$, per-band intensity jitter $\times U[0.8, 1.2]$ with $p = 0.8$, per-band Gaussian noise $\sigma = 0.02$, and per-band z-score. Spectral band dropout, RGB colour jitter, and mixup are deliberately omitted (incompatible with three-band EO + temporal-pair semantics).

3.1 Data

We use the CropNet release of Sentinel-2 imagery [17] for the seven largest U.S. Corn Belt states (Iowa, Illinois, Indiana, Minnesota, Nebraska, Ohio, South Dakota) over 2017–2022, restricted to the growing season (April–September) at biweekly cadence. Each county is partitioned into $\sim$12 grids of 9×9 km, each grid stored as a 224×224 px tile at $\sim$40 m/pixel resolution. We use the agriculture false-colour band combination (B02 Blue, B08 NIR, B11 SWIR1) as a 3-channel input. Total scope: 616 unique counties, 44,140 county-biweek index rows, 852,726 grid tiles, and 3,126 USDA county-year corn-yield labels (the $\sim$12 % of cells with no published yield are small low-acreage counties suppressed for privacy by USDA). Figure 8 (Appendix) lays out the geographic and temporal coverage.

We additionally compute *per-pixel* NDVI from the CropNet B04+B08 quantised reference channels and biweekly-mean NDVI at the county level, used for the hand-crafted baselines in Section 3.4. Figure 9 (Appendix) shows the resulting per-state phenology curves and the global peak-NDVI-vs-yield relationship.

3.2 Geo-SimCLR

The pre-training pipeline is illustrated in Figure 1(a) and an example pretext task on real tiles is shown in Figure 2.

Temporal positive pairs. A SimCLR positive pair is conventionally two stochastic augmentations of the *same* image. We instead construct positives as $(\text{tile}_{c,t,g},\ \text{tile}_{c,t\pm1,g})$ – the same county c and grid index g at adjacent biweeks t and $t \pm 1$. Both tiles are then independently augmented before being fed through the shared encoder. This forces the representation to be invariant to short-term weather, illumination, and atmospheric noise (which differ across consecutive Sentinel-2 passes of the same field) while remaining sensitive to slowly-varying canopy state. After dropping anchors whose county lacks a CRD assignment in the lookup (~ 4.7 %), we obtain 644,922 anchors per epoch.

CRD-balanced anchor sampling. Without intervention, batches drawn uniformly at random over all anchors are dominated by counties from the data-rich, large states (IA, NE), which makes within-batch negatives systematically less climate-diverse. We therefore use a `WeightedRandomSampler` with weights inversely proportional to each anchor's Crop Reporting District frequency ($\mathrm{crd\_id} = \mathrm{state\_ansi} \times 100 + \mathrm{asd\_code}$), yielding 61 unique CRDs across the seven states. Each batch is therefore agro-climatically diversified on average without explicit hard-negative mining.

Safe satellite-aware augmentations. Per view we apply, in order: random horizontal flip ($p = 0.5$), random vertical flip ($p = 0.5$), random rot90 with $k \in \{1, 2, 3\}$ ($p = 0.75$), random crop to side length $\in [160, 208]$ px followed by bilinear resize back to 224 px, random sharpness $\times 2.0$ ($p = 0.5$), per-band intensity jitter $\times U[0.8, 1.2]$ ($p = 0.8$), per-band Gaussian noise $\sigma = 0.02$ in normalised $[0, 1]$ space, and per-band z-score normalisation. We deliberately omit three augmentations that are standard in RGB SimCLR but inappropriate here: spectral band dropout (too destructive on three channels), RGB-flavoured colour jitter (we have no R/G/B channels in the conventional sense), and mixup (would conflate distinct fields and break the temporal-pair semantics).

Architecture and loss. The encoder f is a torchvision ResNet-18 trained from scratch (11.18 M backbone parameters, no input adapter – the default 3-channel 7×7 `conv1` accepts our 3-band tiles directly). The projection head g is a 2-layer MLP $512 \to 256 \to 128$ with BatchNorm + ReLU; its ℓ_2-normalised 128-d output $z = g(f(x))/\|g(f(x))\|_2$ feeds the NT-Xent loss [3],

$$\mathcal{L}_{i\to j} = -\log \frac{\exp(z_i^\top z_j/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k\neq i]} \exp(z_i^\top z_k/\tau)}, \tag{1}$$

with temperature $\tau = 0.2$ and $2N - 1$ in-batch negatives ($N = 1024$). The total parameter count (encoder + projector) is 11.34 M. Training runs for 50 epochs at batch size 1024 across 2 GPUs with AdamW (lr 10^{-3}, weight decay 10^{-4}), a 5-epoch linear warm-up followed by cosine annealing, and AMP mixed precision; full hyperparameters are in Table 5.

3.3 Frozen-feature probe protocol

For downstream evaluation we freeze the pre-trained backbone and aggregate tile features to one vector per (*fips*, *year*) cell by mean-pooling first across grids within each biweek and then across biweeks. Two probes are reported: **Ridge** regression with $\alpha = 1$, and **LightGBM** [14] with 300 trees, learning rate 0.05, num_leaves 31, min_child_samples 5, with 20-round early stopping on a 2021 validation set when applicable.

Split-aware feature choice. A 7-state one-hot indicator and a year scalar can be appended to the encoder features before fitting the probe. We report headline numbers under the per-encoder split-aware choice that wins empirically for *every* encoder: on the time split, Ridge *without* state+year features (a year scalar extrapolated outside the training range hurts the linear model and trees clamp at the boundary leaf, so adding it backfires); on the county split, LightGBM *with* state+year features (year is in-distribution and the trees extract state-year structure cleanly). Table 4 in the appendix records all eight $\{\text{probe}\} \times \{\pm\text{state-year}\} \times \{\text{split}\}$ configurations for full transparency.

Bootstrap CIs. All test-set R^2 and RMSE values are reported with bootstrap 95 % confidence intervals computed from $n = 200$ paired resamples of the held-out $(y_{\text{true}}, \widehat{y})$ pairs, percentiles 2.5/97.5.

3.4 Baselines

We benchmark Geo-SimCLR against five baselines spanning the full chain from hand-crafted indices to large foundation models.

Hand-crafted NDVI (six specifications, Table 2): peak NDVI scalar; growing-season summary (6 features); biweekly NDVI series (12 features); state+year one-hots; state+year + biweekly NDVI (the strongest non-DL specification with 20 features); state+year + peak NDVI. Models are Ridge or LightGBM per the table.

Random ResNet-18. Identical architecture to Geo-SimCLR but with a freshly Kaiming-normal-initialised backbone, never trained. This serves as a sanity baseline: mean-pooled random conv features after BatchNorm capture colour and texture statistics correlated with vegetation.

DINOv2-base [22]: 768-d frozen ViT pre-trained on 142 M natural RGB images. We feed the AG bands as RGB slots $(\mathrm{R}, \mathrm{G}, \mathrm{B}) \leftarrow (\mathrm{B02}, \mathrm{B08}, \mathrm{B11})$ and use the CLS token embedding. 85.7 M parameters.

Prithvi-EO 2.0 [11]: 1024-d frozen ViT pre-trained on Harmonised Landsat-Sentinel imagery, expecting a 6-band HLS input. Since CropNet only provides 3 of the 6 bands (B02, B08, B11), we channel-pad the missing 3 with zeros at inference. 303.9 M parameters. This input mismatch is a documented limitation of the baseline rather than a property of Prithvi itself: the model's effective rank on our channel-padded inputs collapses to $\sim 1.7/1024$ in our diagnostics, severely under-using its capacity.

ConvLSTM [25] as a supervised end-to-end imagery comparator: a per-frame ResNet-18 (random init) producing 512-d per biweek, followed by a single-layer LSTM (hidden 256) and an MLP head with dropout 0.2, trained on z-scored yields with image augmentation applied identically to all 12 biweek frames. Total ∼12.4 M parameters; full hyperparameters in Table 5.

3.5 Splits

We report two complementary generalisation settings. **Time split:** train 2017–2020 / val 2021 / test 2022. This is the strict, no-look-ahead in-season-forecast benchmark: no 2022 tile – not even through positive pairs – enters pre-training. **County split:** a random 20 % of FIPS codes is held out across all six years. This is a spatial generalisation test: the model is asked to predict yield in counties whose imagery was withheld from both pre-training and probe fitting.

4 Empirical results

4.1 Headline encoder comparison

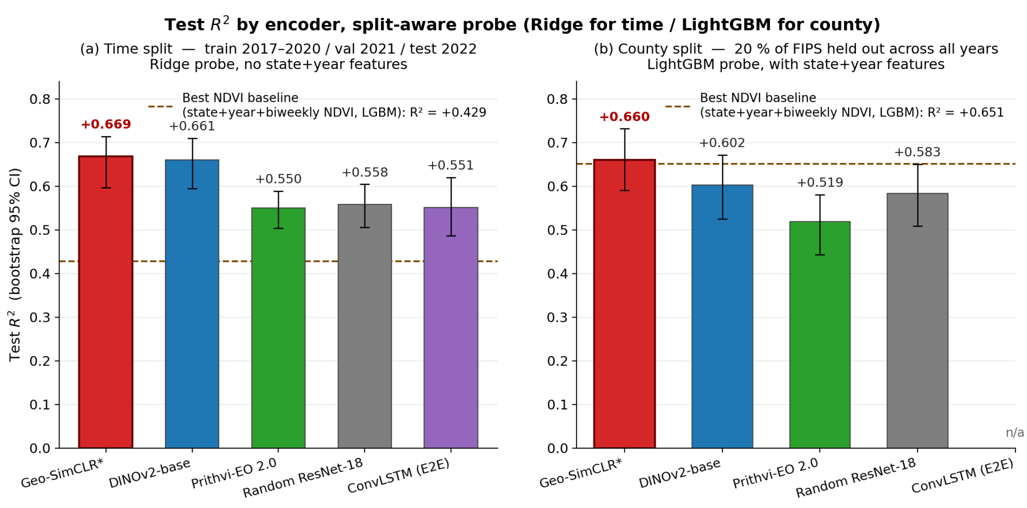

Figure 3: Test R^2 by encoder under the per-encoder split-aware probe choice. Error bars are bootstrap 95 % CIs over $n = 200$ paired resamples; the orange dashed line marks the strongest hand-crafted NDVI baseline (state + year + biweekly NDVI, LightGBM). *Left:* time split; Ridge probe with no state+year features. *Right:* county split; LightGBM probe with state+year features. ConvLSTM is run only on the time split. Geo-SimCLR is the per-panel winner on both splits.

Table 1: Test-set R^2 and RMSE (bushels per acre) by encoder on 7-state US Corn Belt corn yield, under the per-encoder split-aware probe choice. The time-split column uses Ridge with no state+year features; the county-split column uses LightGBM with state+year features (the empirical winner of both {Ridge, LightGBM} $\times$ {with, without state+year} grids for every encoder; see Table 4). Square brackets are bootstrap 95% confidence intervals from $n = 200$ paired resamples of held-out predictions; bold marks the column winner among the deep-learning rows.

		Time split		County split	
Encoder	**Params (M)**	R^2 [95% CI]	**RMSE**	R^2 [95% CI]	**RMSE**
Geo-SimCLR*	11.2	**+0.669 [+0.597, +0.714]**	**19.03**	**+0.660 [+0.591, +0.732]**	**16.60**
DINOv2-base	85.7	+0.661 [+0.594, +0.710]	19.25	+0.602 [+0.525, +0.671]	17.95
Prithvi-EO 2.0	303.9	+0.550 [+0.504, +0.588]	22.17	+0.519 [+0.443, +0.581]	19.74
Random ResNet-18	11.2	+0.558 [+0.506, +0.604]	21.96	+0.583 [+0.509, +0.650]	18.37
ConvLSTM (E2E)	12.4	+0.551 [+0.487, +0.620]	21.95	–	–
Best NDVI baseline[a]	–	+0.429	24.78	+0.651	18.15

* Our model. [a] Best NDVI baseline = state one-hot + year + 12 biweekly NDVI features, LightGBM (strongest of six tabular NDVI specifications; see Table 2). ConvLSTM is reported only on the time split.

Geo-SimCLR is the strongest single encoder on both splits (Table 1, Figure 3). On the time split it reaches test $R^2 = 0.669$ [0.597, 0.714] with RMSE 19.0 bu/ac, beating the strongest non-DL baseline by $+0.24\,R^2$ and the supervised end-to-end ConvLSTM by $+0.118\,R^2$ – a clean win in the "frozen-features-and-linear-probe beats supervised end-to-end" direction. On the county split Geo-SimCLR reaches $R^2 = 0.660$ [0.591, 0.732] with RMSE 16.6 bu/ac. Here the strongest tabular baseline (state+year + biweekly NDVI LightGBM) is closer at $R^2 = 0.651$, because year and state are in-distribution for the county split and absorb a large fraction of the yield variance independently of imagery; the imagery-driven margin is therefore narrow ($+0.009\,R^2$) but consistent across both Ridge and LightGBM probe choices.

Probe-choice consistency. The split-aware probe pattern is not cherry-picked to favour one encoder. As Table 4 (appendix) documents, Ridge beats LightGBM on the time split for every one of the four DL encoders, and LightGBM with state+year features beats every alternative on the county split for every encoder. DINOv2-base narrowly outranks Geo-SimCLR on a handful of off-headline cells (most notably time-split / LightGBM / +state-year, by $\sim+0.04\,R^2$); Geo-SimCLR wins or ties on the empirically-best cell for every encoder.

4.2 Hand-crafted NDVI baselines

The six NDVI specifications (Table 2) reproduce well-known patterns: the biweekly-NDVI series (12 features) substantially outperforms the peak-NDVI scalar on both splits (e.g. R^2 0.475 vs. 0.261 on the county split), capturing the canopy phenology that a single August snapshot misses; and adding state+year one-hots yields the strongest non-DL specification on both splits ($R^2 = 0.429$ time, 0.651 county). The county-split numbers are systematically larger than the time-split numbers because year-to-year variance (driven by weather and yield trend) exceeds across-county variance for fixed years, and the time split asks the model to extrapolate to a year it has never seen.

Table 2: Hand-crafted NDVI baselines for 7-state corn-yield prediction (LightGBM and Ridge), test-set R^2 and RMSE (bushels per acre) on both splits. Bold values are the per-column winner.

			Time split		County split	
Specification	**#feats**	**Model**	R^2	**RMSE**	R^2	**RMSE**
Peak NDVI	1	Ridge	+0.141	30.38	+0.261	26.43
Growing-season summary	6	LightGBM	+0.200	29.33	+0.363	24.54
Biweekly NDVI series	12	LightGBM	+0.279	27.85	+0.475	22.27
State + year (no NDVI)	8	Ridge	+0.199	29.34	+0.285	25.85
State + year + biweekly NDVI	20	LightGBM	**+0.429**	**24.78**	**+0.651**	**18.15**
State + year + peak NDVI	9	Ridge	+0.235	28.68	+0.427	23.26

4.3 Interpreting the Encoder Ranking

The headline performance ranking in Table 1 — Geo-SimCLR $\succ$ DINOv2-base $\succ$ ConvLSTM $\sim$ Random ResNet-18 $\succ$ Prithvi-EO 2.0 — highlights several key representation learning dynamics.First, Geo-SimCLR's significant margin over the supervised end-to-end ConvLSTM ($+0.118\,R^2$ on the time split) underscores the data efficiency of contrastive learning. Pre-training on $\sim$852,k unlabelled tiles avoids the severe overfitting that occurs when training deep networks end-to-end on just $\sim$2,100 annual yield labels.Geo-SimCLR also outperforms the much larger, EO-specialised Prithvi-EO 2.0, primarily due to an input mismatch. Prithvi expects a 6-band HLS input; zero-padding our 3-band CropNet data causes its effective rank to collapse to $\sim 1.7/1024$, severely underutilising its capacity. Conversely, DINOv2-base nearly matches Geo-SimCLR despite being pre-trained on natural RGB images. Its robust CLS token transfers well to agriculture false-colour inputs, though Geo-SimCLR maintains a slight edge because its temporal pretext task is domain-aware and avoids natural-image biases.Finally, a randomly initialised ResNet-18 achieves an $R^2 > 0$ simply by capturing basic colour and texture statistics correlated with vegetation. Therefore, Geo-SimCLR's true representation learning gain is best quantified not by its absolute score, but by its margin over this random baseline ($+0.111\,R^2$ on the time split and $+0.077\,R^2$ on the county split).

4.4 Self-supervised training trajectory

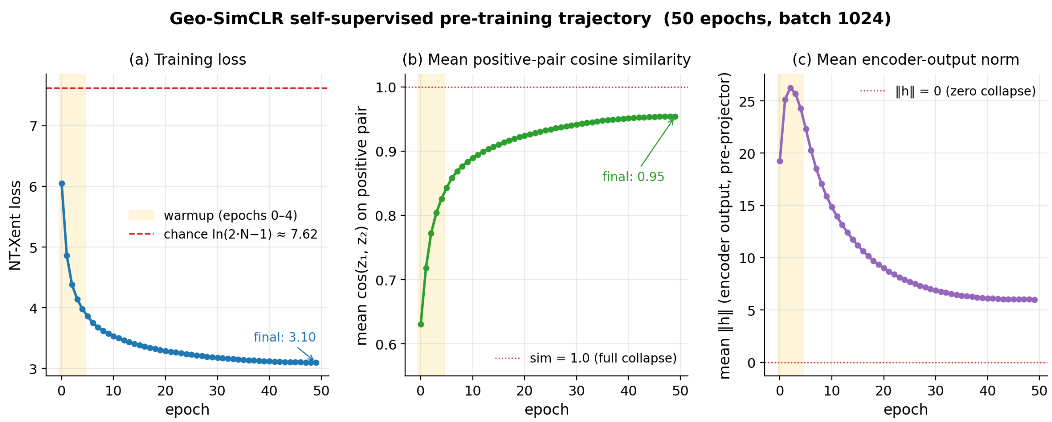

Figure 4: Geo-SimCLR pre-training trajectory over 50 epochs at batch 1024. (a) NT-Xent loss falls cleanly from 6.05 at epoch 0 (already below the chance level $\ln(2N-1) \approx 7.62$) to 3.10 at epoch 49. (b) Mean positive-pair cosine similarity rises monotonically to $\sim$0.95 – a warning that adjacent-biweek temporal positives are inherently easy (Section 4.6). (c) Mean encoder-output norm $\|h\|$ is stable, ruling out the standard zero-collapse failure mode in which the encoder maps everything to the origin. The shaded yellow band marks the 5-epoch linear LR warm-up.

The training trajectory in Figure 4 shows clean, non-pathological optimisation: loss converges from chance, and the encoder output norm remains bounded. The one early warning sign is the monotone climb of the mean positive-pair cosine similarity to $\sim$0.95. A healthy SimCLR training trajectory tends to plateau much lower (typically 0.5–0.8 on natural images). However, the similarity starts at aroun 0.6, so the proximity to 1 somewhat less surprising, but it does suggest the temporal-positive pretext task is too easy: the encoder can satisfy NT-Xent by simply collapsing same-county-and-near-time pairs onto each other rather than learning richer fine-grained features. We diagnose this collapse quantitatively next.

4.5 What did the encoder learn?

The M-PHATE embedding (Figure 5) has two complementary readings. State coloring – the (a) panel – produces clean, geographically-coherent clusters: counties within the same state map to a common region of the embedding, and adjacent states (IL/IN) cluster more tightly with each other than with distant states (SD/MN). This is the *positive* side of the story: the encoder reliably encodes the agro-climatic identity of each county. Biweek coloring – the (b) panel – shows the reverse pattern: biweeks blend smoothly within each state cluster rather than tracing distinct phenological trajectories. M-PHATE's inter-slice kernel imposes some temporal coherence by construction, so this is a generous upper bound; the next subsection quantifies how much phenology actually survives.

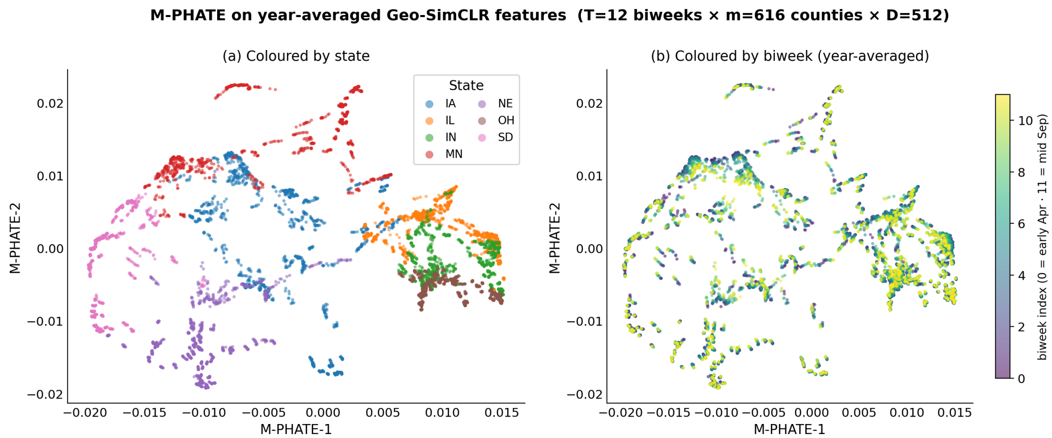

Figure 5: M-PHATE [7] 2-D embedding of Geo-SimCLR features on the year-averaged tensor (12 biweeks $\times$ 616 counties $\times$ 512 features), with intra-slice $k = 5$ and inter-slice $k = 6$ neighbour kernels. (a) Coloured by state: state-coherent clusters separate cleanly. (b) Coloured by biweek index: biweek transitions are smooth but tightly overlapping within each state cluster, indicating that seasonal phenology is only weakly encoded relative to geographic identity. M-PHATE's inter-slice kernel imposes some temporal coherence by construction, so this is a generous upper bound on what the encoder learned about phenology.

4.6 Potential collapse diagnostic

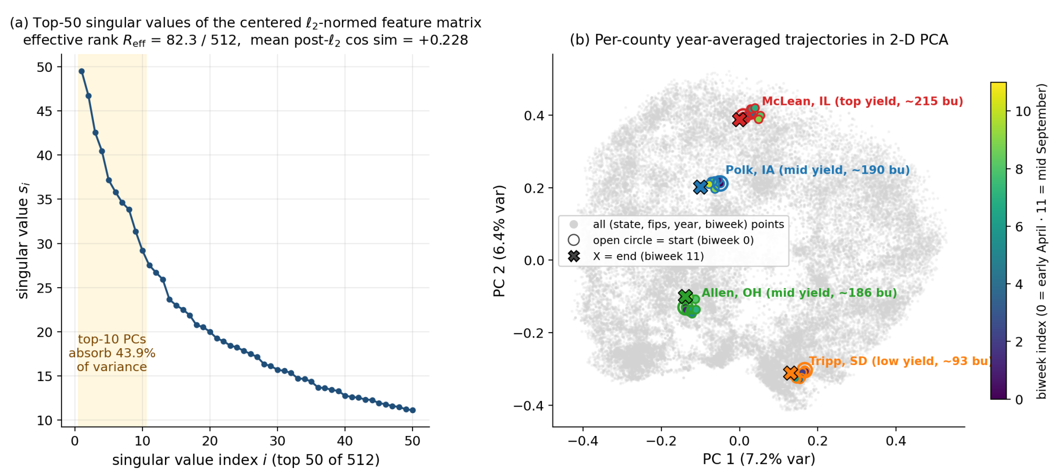

Figure 6: Dimensional-collapse diagnostic on the full ℓ_2-normalised pooled feature matrix. (a) Top-50 singular values of the centered feature matrix; the shaded region marks the top-10 PCs, which already absorb 43.9 % of the variance, and the effective rank $R_{\text{eff}} = \exp(-\sum_i p_i \log p_i)$ of Roy and Vetterli [23] is 82.3 out of 512. (b) Per-county year-averaged trajectories in 2-D PCA for four hand-picked counties spanning the yield range; each county's 12 biweek points span only 4–6 % of the global pairwise distance, so the seasonal trajectory is compressed almost to a single point relative to the across-county spread.

Table 3: Geo-SimCLR representation diagnostics on the full 7-state pooled feature matrix ($n = 44{,}140$ tile vectors $\times$ 512 dimensions, ℓ_2-normalized to match the cosine NT-Xent geometry). Effective rank follows Roy and Vetterli (2007); cosine-similarity statistics are computed over $n = 2{,}000$ random tile-vector pairs.

Diagnostic	**Value**
Number of pooled tile vectors	44,140
Embedding dimensionality D	512
Effective rank $R_{\text{eff}} = \exp(-\sum_i p_i \log p_i)$	82.3 / 512
Top-1 PC variance fraction	7.21%
Top-10 PC variance fraction	43.92%
Top-30 PC variance fraction	69.14%
Singular-value ratio s_0/s_1	1.06
Singular-value ratio s_0/s_{10}	1.80
Within-county feature variance	2.21e-04
Between-county feature variance	1.29e-03
Variance ratio (within / between)	0.171
Mean post-ℓ_2 cosine similarity (random pairs)	+0.228 $\pm$ 0.137

The SVD spectrum and per-county PCA trajectories in Figure 6 provide quantitative confirmation of the qualitative M-PHATE finding. Three numbers tell the story (Table 3). First, the effective rank of the pooled, ℓ_2-normalised feature matrix is $82.3/512$: the encoder concentrates the data on a low-dimensional subspace using $\sim$16 % of its nominal capacity. Second, the mean cosine similarity between random pairs of ℓ_2-normalised tile vectors is $+0.228$, far above the ≈ 0 expected from a healthy SimCLR encoder [3, 12]. Third, the within-county feature variance is only 17 % of the between-county feature variance: most of the information in the embedding is which county the tile is from, not when in the season it was taken or how that field is doing.

The PCA trajectories in Figure 6(b) make this concrete: each county's 12 biweek points (early-April open circle through mid-September X marker) span only 4–6 % of the global pairwise distance. A county imaged in April and the same county imaged in September map to almost the same point in the encoder's representation. This is the canonical signature of dimensional collapse onto a single dominant direction (here: county identity). The appendix figure (Figure 10) corroborates the same finding through plain PHATE and UMAP, both of which produce ball-shaped embeddings with significant state and biweek mixing.

We hypothesise the proximate cause is the same one Figure 4(b) flagged: same-county tiles separated by only ± 14 days at 9 km field scale are intrinsically very similar, and the contrastive objective can satisfy NT-Xent by reinforcing static county identity rather than learning fine-grained phenology. Section 5 discusses concrete fixes.

4.7 Pixel attribution via GradCAM

The GradCAM maps in Figure 7 show the regions of each tile that drive the encoder's contribution to the Ridge yield head. The pattern is consistent across both yield extremes: attention concentrates on cropland regions (the vivid green areas in agriculture false-colour) and de-emphasises water bodies, urban pixels, and partially cloudy regions. Given the potential collapse, these maps should be interpreted as *cropland-vs-non-cropland attribution* – the encoder tries to focus on where in each tile the agriculture is – rather than as a per-pixel readout of how that crop is currently doing.

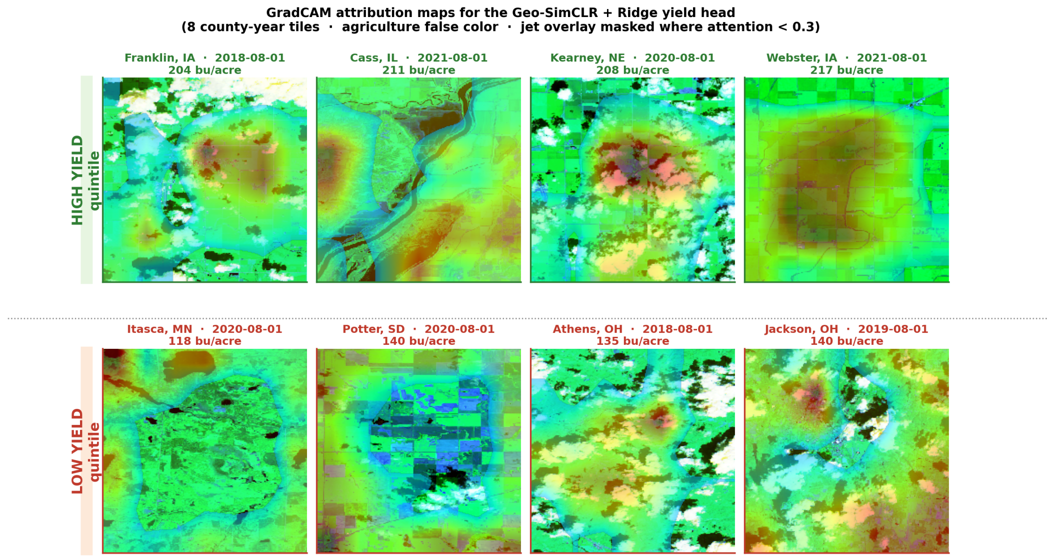

Figure 7: GradCAM [24] attribution maps for the composition of the frozen Geo-SimCLR encoder and a Ridge yield head, computed via Captum's `LayerGradCam` [16] on `layer4[-1]`. Top row: four highest-yield-quintile county-years; bottom row: four lowest. Tiles are rendered in agriculture false-colour with a global 2nd–98th-percentile contrast stretch; the jet overlay is masked to the top 70 % of attention so only high-attention regions are coloured.

5 Discussion, limitations, and future work

Summary. Geo-SimCLR – a SimCLR variant with temporal positive pairs, CRD-balanced anchor sampling, and safe satellite-aware augmentations – is the strongest single encoder on both temporal and spatial generalisation splits of seven-state Corn Belt corn-yield prediction, beating frozen foundation models that have one to two orders of magnitude more parameters and beating supervised end-to-end ConvLSTM at the same input modality. The margin over the strongest hand-crafted NDVI baseline is large on the time split ($+0.24\, R^2$) and narrow but consistent on the county split ($+0.009\, R^2$).

Limitations. The learned representation exhibits partial dimensional collapse along county-identity directions (effective rank $82/512$, mean post-ℓ_2 cosine similarity $+0.228$, within/between county variance ratio 0.17). M-PHATE and per-county PCA trajectories show that geographic structure is encoded cleanly while seasonal phenology is only weakly encoded; the GradCAM maps should therefore be read as static cropland-vs-non-cropland evidence rather than as per-pixel phenology attribution. The 11.2 M-parameter ResNet-18 backbone is small by SSL-paper standards and may itself cap the achievable representation quality; the seven states and six years restrict generalisation claims.

Future work. We see eight concrete next steps, in priority order.

1. **Harder positives and stronger spatial augmentations.** Same-field pairs at $\pm$3–5 biweeks (or cross-year same-county pairs) and tighter crop ranges should make the contrastive task non-trivial and directly attack the easy-positives root cause.
2. **Decorrelation regularisation.** Adding a Barlow-Twins [27] or VICReg [2] off-diagonal cross-correlation term to the NT-Xent loss should encourage the representation to spread across the full 512 dimensions rather than collapsing onto county identity.
3. **Head-to-head against the temporal-positive SSL lineage.** Benchmarking against SeCo [18], GASSL [1], GeoSSL, and SatMAE [4] under the same probe protocol would position Geo-SimCLR within its closest prior art.

4. **Larger backbone** (ResNet-50, ConvNeXt, ViT-S) – contrastive SSL is well-known to scale with parameters, and our 11.2 M is small relative to recent literature.
5. **4-channel AG+NDVI input.** Explicitly providing NDVI as a fourth channel removes the burden of rediscovering chlorophyll structure from B02/B08/B11.
6. **Partial fine-tuning** (last residual block + head, low backbone LR) on yield – the canonical SSL evaluation protocol – is expected to close any remaining gap to supervised end-to-end models.
7. **USDA Weekly Crop Progress and Condition** ($\sim$35 k state-week labels) is a substantially denser supervision signal than annual yield and is a natural fine-tuning target.
8. **Wider scope**: more states, more crops (corn $\leftrightarrow$ soybean transfer, cotton, winter wheat), weather ablation against WRF-HRRR, and eventually cross-country generalisation.

Broader applicability. The recipe – temporal positive pairs from co-located satellite scenes, climate-zone-balanced anchor sampling, and a safe satellite-aware augmentation set – is not specific to corn yield. Better domain-tailored representations of medium-resolution multispectral imagery have downstream value across climate-change attribution, deforestation and biodiversity monitoring, urban-growth analysis, water-resource and drought tracking, and disaster response, all of which share the same structural property: abundant unlabelled imagery and scarce, expensive ground-truth labels. Thus, we believe this literature has a lot to develop and hope that this analysis contributes to the satellite imagery representation research development.

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A Supplementary material

A.1 Data scope and NDVI phenology

Figures 8 and 9 show the dataset scope and per-state NDVI phenology that motivate the hand-crafted baseline specifications in Section 3.4.

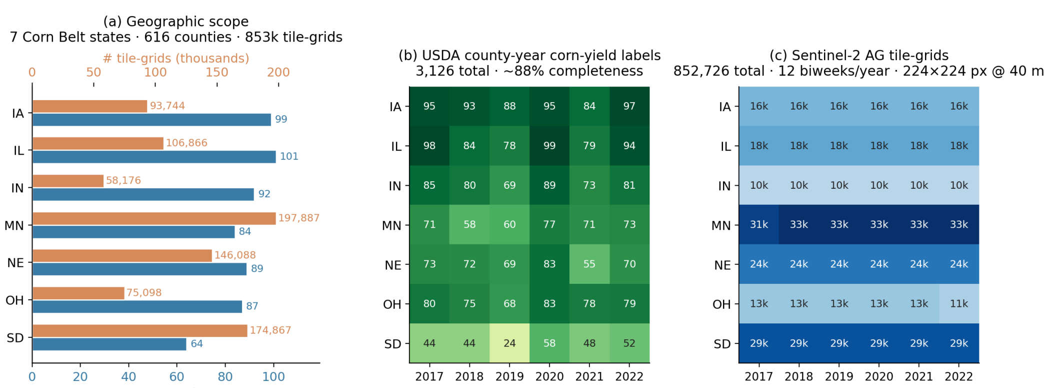

Figure 8: Data scope. (a) Per-state county and tile-grid counts: 7 states, 616 counties, 852,726 tile-grids in total. (b) USDA county-year corn-yield label counts (3,126 total, ∼88 % completeness; cells without labels are small low-acreage counties suppressed by USDA for privacy). (c) Sentinel-2 AG tile-grid counts per (state, year), 12 biweeks per year.

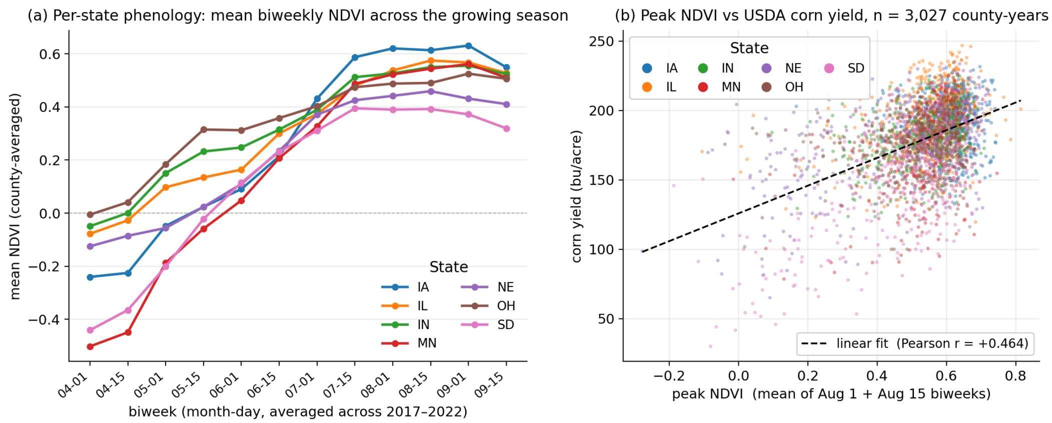

Figure 9: Per-pixel NDVI computed from CropNet's quantised B04 + B08 reference channels. (a) Per-state mean biweekly NDVI across the growing season (averaged over 2017–2022); the canonical green-up $\rightarrow$ Aug peak $\rightarrow$ senescence pattern is visible in all seven states, with MN/SD starting notably negative in early April due to lingering snow and wet bare soil. (b) Per-county-year peak NDVI (mean of Aug 1 + Aug 15 biweeks) vs. USDA corn yield, coloured by state, with the global linear fit and Pearson r overlaid. Within-state correlations are higher than the global $r = +0.46$.

A.2 Full encoder $\times$ probe grid

Table 4: Full encoder $\times$ probe $\times$ split $\times$ {with, without} state+year features grid for the four deep-learning encoders (R^2, higher is better). $\pm$SY denotes whether the 7-state one-hot and the year scalar are appended to the encoder features before fitting the probe. Bold values are the per-column winner; an asterisk (*) marks our model. RMSE versions omitted for compactness.

	Time split				**County split**			
	Ridge		LGBM		Ridge		LGBM	
Encoder	−SY	+SY	−SY	+SY	−SY	+SY	−SY	+SY
Geo-SimCLR*	**+0.669**	+0.548	+0.624	+0.542	+0.451	+0.452	**+0.596**	**+0.660**
DINOv2-base	+0.661	**+0.609**	**+0.641**	**+0.582**	**+0.541**	**+0.559**	+0.495	+0.602
Prithvi-EO 2.0	+0.550	+0.573	+0.424	+0.483	+0.358	+0.391	+0.282	+0.519
Random ResNet-18	+0.558	+0.565	+0.506	+0.560	+0.312	+0.354	+0.404	+0.583

A.3 Plain PHATE and UMAP corroboration of the collapse

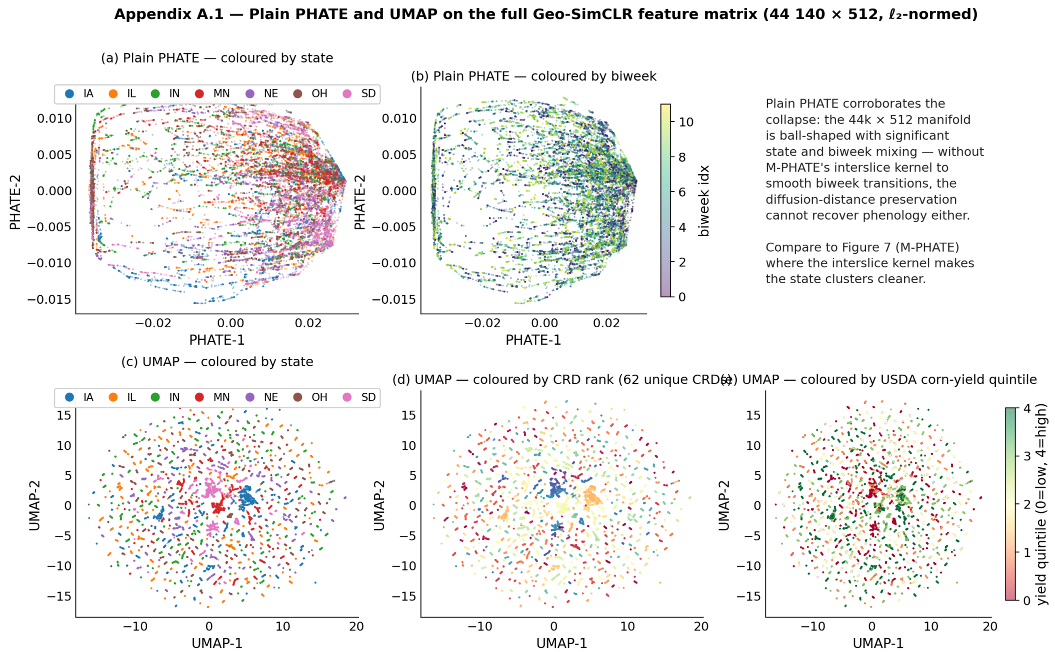

Figure 10: Plain PHATE [20] and UMAP [19] on the same 44,140 $\times$ 512 ℓ_2-normalised feature matrix used for the collapse diagnostic in Section 4.6. (a, b) PHATE coloured by state and biweek; without M-PHATE's inter-slice kernel, the diffusion-distance preservation cannot recover phenology either, and the embedding is ball-shaped with significant state and biweek mixing. (c, d, e) UMAP coloured by state, by Crop Reporting District rank (62 unique CRDs), and by USDA corn-yield quintile. All three views corroborate the SVD-based collapse evidence: the manifold is too dense for UMAP's local-distance focus to find clean state boundaries.

A.4 Full hyperparameter table

Table 5: Full hyperparameter and training-protocol summary for Geo-SimCLR self-supervised pre-training, the supervised end-to-end ConvLSTM baseline, and the frozen-feature probe protocol. Values are exactly those used to produce the headline results in Table 1.

Geo-SimCLR self-supervised pre-training (notebook 04)	
Backbone	ResNet-18 from scratch (3-channel input, no adapter)
Projection head	MLP $512 \to 256 \to 128$ with BN+ReLU; ℓ_2-normalized output
Total parameters	$11.34\,\mathrm{M}$ (encoder $11.18\,\mathrm{M}$ + projector $0.16\,\mathrm{M}$)
Loss	NT-Xent on ℓ_2-normalized 128-d projections
Temperature τ	0.20
Batch size	1024 (effective; DataParallel across 2 GPUs)
Optimizer	AdamW, $\mathrm{lr} = 10^{-3}$, $\mathrm{wd} = 10^{-4}$
Scheduler	LinearLR warmup ($\mathrm{lr}/100 \to \mathrm{lr}$) over 5 epochs, then CosineAnnealingLR ($\eta_{\min} = 10^{-6}$) for 45 epochs
Epochs	50
Mixed precision	AMP (bfloat16 autocast) + GradScaler
Anchors per epoch	$\sim 644{,}922$ after dropping NaN-CRD anchors
CRD-balanced sampling	WeightedRandomSampler, weights $\propto 1/(\#\text{anchors per CRD})$, 61 CRDs
Augmentations (per view)	hflip 0.5; vflip 0.5; rot90 $k \in \{1, 2, 3\}$ p=0.75; random crop 160–208 px $\to$ resize 224; sharpness $\times 2.0$ p=0.5; per-band intensity $\times U[0.8, 1.2]$ p=0.8; per-band Gaussian noise $\sigma = 0.02$; per-band z-score
Excluded augmentations	spectral band dropout, RGB color jitter, mixup
Data scope	7 states (IA, IL, IN, MN, NE, OH, SD), 6 years (2017–2022), Apr–Sep biweekly, 224×224 px AG (B02, B08, B11) tiles
Wall time	∼5–7 min / epoch × 50 ep ≈ 5 h on 2 GPUs
ConvLSTM end-to-end supervised baseline (notebook 05)	
Per-frame encoder	ResNet-18 (random init) producing 512-d per biweek
Recurrence	single-layer LSTM, hidden 256
Head	MLP $256 \to 64 \to 1$, dropout 0.2
Target	z-scored on training yields
Optimizer	AdamW, $\mathrm{lr} = 10^{-4}$, $\mathrm{wd} = 10^{-4}$
Scheduler	CosineAnnealingLR ($T_{\max} = 30$, $\eta_{\min} = 10^{-6}$)
Batch size / Epochs	32 / 30; checkpoint = best 2021-validation RMSE
Augmentations	identical hflip / vflip / rot90 / random crop applied to all 12 biweek frames per example
Frozen-feature probe protocol (notebook 05, all encoders)	
Pooling	mean-pool tile features per (fips, year, biweek), then per (fips, year)
Linear probe	Ridge, $\alpha = 1$ (no CV), no normalization
Tree probe	LightGBM: 300 trees, lr 0.05, num_leaves 31, min_child_samples 5
LGBM early stopping	20 rounds on 2021 validation set (time split only)
State+year features	7-state one-hot + year scalar appended when applicable
Splits	*Time*: train 2017–2020 / val 2021 / test 2022. *County*: random 20% FIPS held out, all years.
Bootstrap CIs	$n = 200$ paired resamples of held-out predictions, percentiles 2.5/97.5

A.5 Compute footprint

Total wall time for the headline pipeline (data acquisition, baselines, self-supervised pre-training, feature extraction, downstream probes, and visualisation) was approximately 14 hours on a Yale High-Performance Computing node with 18 CPU cores, 24 GiB of RAM, and 2 GPUs. Self-supervised pre-training itself was the dominant cost at ∼19 hours (50 epochs × ∼12 minutes/epoch with AMP and DataParallel across 2 GPUs). Feature extraction for the four encoders took ∼2 hours; probe fitting and visualisation ran in ∼1 hour total on a local CPU.